Section 2

What is the key difference between “thinking rationally” and “thinking humanly” in AI?

- Thinking rationally involves imitating human behavior
- Thinking humanly follows strict logical rules for decision-making
- Thinking rationally is based on formal logic, while thinking humanly aims to replicate human thought processes
- Thinking humanly is more computationally efficient than thinking rationally

In the PEAS framework for an autonomous taxi, which of the following is an actuator?

- GPS system
- Speedometer
- Steering mechanism
- Road condition detector

The A* algorithm expands nodes in order of:

- Their depth in the search tree
- Their heuristic value h(n)
- Their path cost g(n)
- Their total estimated cost f(n) = g(n) + h(n)

In uniform-cost search, nodes are expanded based on:

- The heuristic value h(n)
- The depth of the node
- The cost from the start node (g(n))
- The total estimated cost (g(n) + h(n))

Why is intelligence difficult to define?

- It is a purely biological trait
- It consists of multiple abilities such as reasoning, learning, and problem-solving
- It is only relevant to human cognition
- It can be measured only through IQ tests

What is one of the limitations of the “Thinking Humanly” approach to AI?

- It does not attempt to understand human cognition
- It ignores human biases
- It requires AI to replicate complex neurological processes
- It relies solely on probabilistic models

Which of the following is an advantage of Iterative Deepening Search over Breadth-First Search?

- Uses less memory
- Guarantees finding the shortest path
- Faster execution time
- Avoids loops in the search space

Which of the following is a goal-based agent likely to use?

- A predefined set of actions
- A utility function
- A search algorithm
- A reactive behaviour system

If a problem has multiple goal states, which search strategy is most effective?

- Depth-First Search
- Breadth-First Search
- Bidirectional Search
- Iterative Deepening Search

Which of the following is true about Greedy Best-First Search?

- It guarantees finding the shortest path
- It is incomplete and not optimal
- It always finds the most efficient solution
- It never gets stuck in infinite loops

Why did early AI programs like General Problem Solver (GPS) struggle with scalability?

- They used weak methods that could not handle complex problems efficiently
- They required too much human supervision
- They were limited to physical robots
- They relied on neural networks, which were not well developed at the time

A vacuum cleaner robot that follows simple predefined rules without learning is an example of a:

- Model-based agent
- Reflex agent
- Learning agent
- Utility-based agent

If a heuristic function overestimates the true cost to reach the goal, then:

- The search will still be optimal
- The algorithm will expand fewer nodes
- The search may fail to find the optimal solution
- The search will become exhaustive

Which of the following best describes a utility function?

- A function that assigns a numerical value to different states based on desirability
- A function that maps sensory input to actions
- A simple rule-based decision-making system
- A method for predicting future states of the environment

What is the worst-case time complexity of Depth-First Search in a graph with branching factor b and depth d?

- O(b^d)
- O(d^b)
- O(bd)
- O(d^2b)

Section 1

which definition of AI focuses on machines behaving in ways that humans consider intelligent?

- thinking rationally
- acting humanly
- acting rationally
- thinking logically

what is the space complexity of A* search in the worst case?

- O(d^b)
- O(b^d)
- O(d^2b)
- O(bd)

why is the “Laws of Thought” approach to AI challenging?

- many real-world problems cannot be expressed in formal logic
- it is the only valid AI approach
- it ignores reasoning and decision making
- it only applies to biological intelligence

what is a rational agent expected to do in a stochastic environment?

- ignore uncertainty and act optimally
- consider probabilities and expected outcomes
- always repeat the same action for consistency
- act randomly to explore possibilities

what is a major drawback of breadth first search?

- it is not complete
- it uses excessive memory
- it cannot handle negative costs
- it does not guarantee a solution

the best first search algorithm selects the next node to expand based on:

- the depth of the node
- the total path cost g(n)
- the heuristic function h(n)
- the total path cost g(n) + the heuristic function h(n)

which search strategy is guaranteed to be optimal only if the cost function is a non-decreasing function of the depth of the node?

- depth first search
- breadth first search
- uniform cost search
- greedy best first search

what problem does NP-completeness highlight in AI?

- some problems are too hard to compute efficiently
- AI can never surpass human intelligence
- logical reasoning is impossible
- AI cannot be applied to decision-making

what is a key distinction between “acting rationally” and “acting humanly” in AI?

- acting rationally aims for optimal decisions, while acting humanly focuses on mimicking human behavior
- acting rationally ignores decision making
- acting humanly is always superior
- acting humanly leads to perfect AI performance

in A* search, which condition must the heuristic function satisfy to ensure optimality?

- it must be greater than the actual cost
- it must be an underestimate of the actual cost
- it should at most be constant with respect to the actual cost
- it must be a never-decreasing random value with respect to the actual cost

the difference between A* and greedy best first search is that A* considers:

- only the heuristic value
- the path cost and the heuristic together
- the maximum depth of the tree
- only the branching factor

a heuristic function is considered consistent if it satisfies which condition?

- $h(n) \leq h^*(n)$
- $h(n) \leq c(n,n') + h(n')$
- $h(n) \geq h^*(n)$
- $h(n) = 0$ for all nodes

which of the following is NOT a characteristic of an optimal search algorithm?

- completeness
- optimality
- infinite branching factor
- efficiency

what is a major challenge in defining intelligence?

- intelligence is only about logical reasoning
- intelligence involves multiple attributes that are difficult to measure
- intelligence is purely about acting humanly
- intelligence can only be measured using the Turing Test